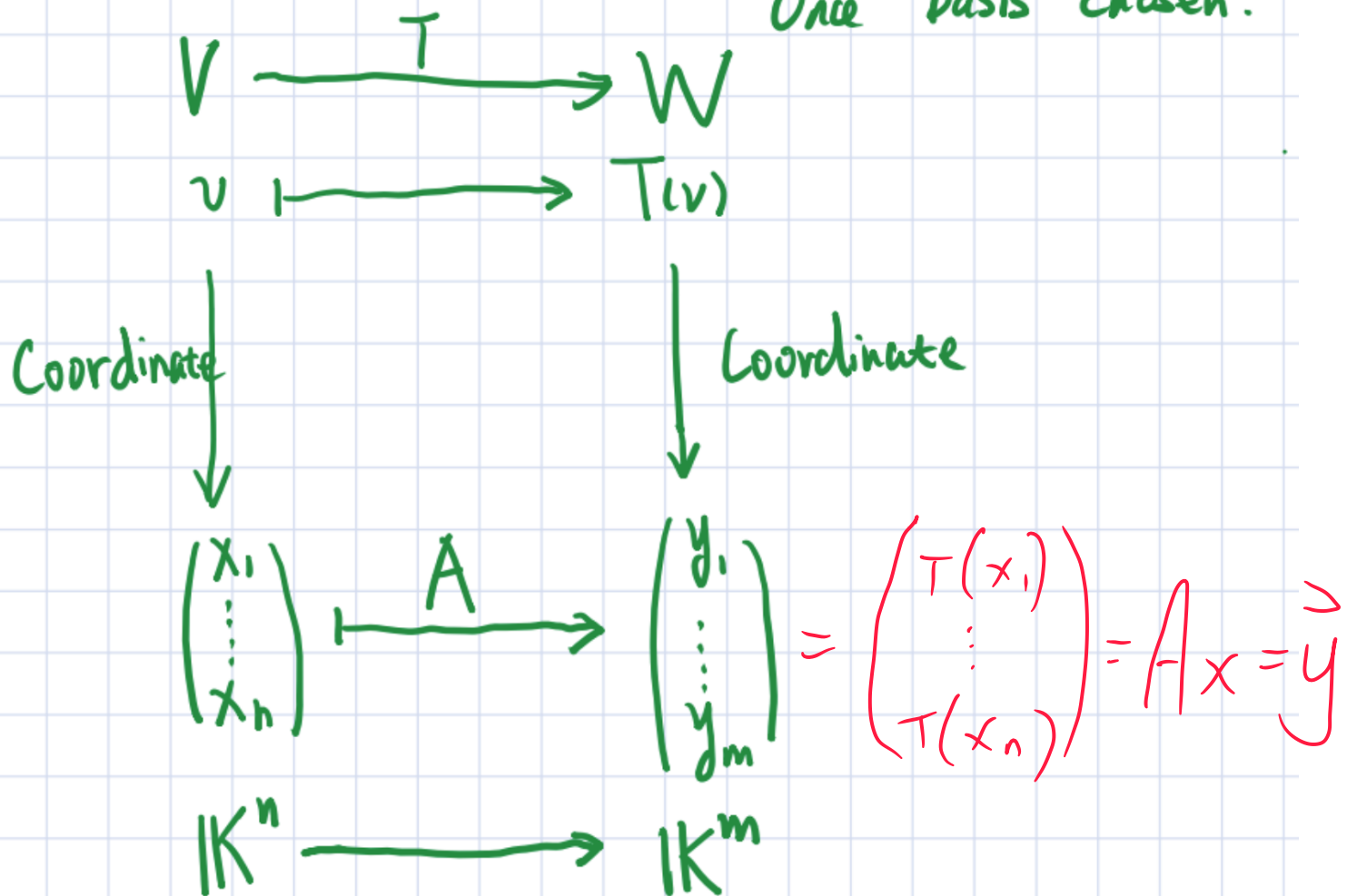


Once basis chosen:



$$\vec{v} = c_1 \vec{\beta}_1 + c_2 \vec{\beta}_2 + \dots + c_n \vec{\beta}_n$$

$\vec{\beta}_i$ basis vectors

c_i coordinate vectors

$\vec{v}$ position vector

$$T(\vec{v}) = T(c_1 \vec{\beta}_1 + \dots + c_n \vec{\beta}_n)$$

$$= c_1 T(\vec{\beta}_1) + \dots + c_n T(\vec{\beta}_n)$$

C_i coordinates don't change in linear mapping
 T linear mapping transforms basis vectors

$[\vec{v}]_B$ is coordinate vector under basis B for position vector $\vec{v}$

$[x_1, \dots, x_n]^T$ above is $[\vec{v}]_B = [c_1, \dots, c_n]$

For change of basis from B to C

$$[\vec{v}]_C = [P]_{C \leftarrow B} [\vec{v}]_B$$

new coord change of Basis matrix old coord

$\swarrow$ B_i basis vectors of B

$$[P]_{C \leftarrow B} = \begin{bmatrix} [B_1]_C & [B_2]_C & \dots & [B_n]_C \end{bmatrix}$$

So in new basis both coord & basis vec change

$$\vec{v} = c_1 B_1 + \dots + c_n B_n = d_1 \gamma_1 + \dots + d_n \gamma_n$$

pos. vector in basis B in basis C

$$\vec{v} = \underbrace{B}_{[B_1 \ B_2 \ \dots \ B_n]} [\vec{v}]_B = \underbrace{\gamma}_{[\gamma_1 \ \gamma_2 \ \dots \ \gamma_n]} [\vec{v}]_C$$

$$B = \gamma [P]_{C \leftarrow B}$$

basis matrix B basis matrix C transition matrix B to C
 can only act on C-coord.

$$\gamma = B [P]_{B \leftarrow C}$$

transition C to B

behavior of change of Basis P
 changes depending on whether it acts
 on coordinates:

$$[v]_C = [P]_{C \leftarrow B} [v]_B$$

acts on B to get C

or on bases directly

$$B = \gamma [P]_{C \leftarrow B} \in \text{acts on } C \text{ to get } B$$

$$\vec{v} = \underbrace{\gamma \cdot [P]_{C \leftarrow B}}_B \cdot \overbrace{[\vec{v}]_C}^{[\vec{v}]_B} = B [\vec{v}]_B$$

- Change of Basis for a Linear map:

Domain: V Bases $\mathcal{E} = (e_1, \dots, e_n)$ and $\mathcal{E}' = (e'_1, \dots, e'_n)$

Codomain: W Bases $\mathcal{F} = (f_1, \dots, f_m)$ and $\mathcal{F}' = (f'_1, \dots, f'_m)$

Change of Basis: $(e'_1, \dots, e'_n) = (e_1, \dots, e_n) P$ (in V)
 $(f'_1, \dots, f'_m) = (f_1, \dots, f_m) Q$ (in W)

Acts from right to go to prime?

P invertible $n \times n$; Q invertible $m \times m$

Take $v \in V$: Coord. vector $[v]_{\mathcal{E}} = P [v]_{\mathcal{E}'} \leftarrow$

Acts from left to send back to original basis?

$$\mapsto v = (e_1, \dots, e_n) [v]_{\mathcal{E}} = (e'_1, \dots, e'_n) [v]_{\mathcal{E}'}$$

$$= (e_1, \dots, e_n) P [v]_{\mathcal{E}'}$$

Similarly $[w]_{\mathcal{F}} = Q [w]_{\mathcal{F}'}$ for any $w \in W$.

Let A be the matrix of T from $\mathcal{E} \rightarrow \mathcal{F}$

B " " " " " from $\mathcal{E}' \rightarrow \mathcal{F}'$

Then: $B = Q^{-1} A P$

P is $[P]_{\mathcal{E} \leftarrow \mathcal{E}'}$ or $[P]_{\mathcal{E}' \leftarrow \mathcal{E}}$
 depending on position

When to use P^{-1}
 vs act from right vs left?

Then: $B = Q^T A P$.

To see this: for every $v \in V$.

$$[T(v)]_{\mathcal{F}} = A[v]_{\mathcal{E}} = AP[v]_{\mathcal{E}'} \quad (*) \quad \leftarrow$$

On the other hand, do a change of basis for $T(v)$

$$[T(v)]_{\mathcal{F}} = Q[T(v)]_{\mathcal{F}'} = QB[v]_{\mathcal{E}'} \quad (**) \quad \leftarrow$$

Equate (*) and (**):

$$AP[v]_{\mathcal{E}'} = QB[v]_{\mathcal{E}'}.$$

This holds for all vectors $[v]_{\mathcal{E}'}$ that is

all vectors in K^n then $AP = QB$

As Q is invertible: $B = Q^T A P$.

$$\begin{matrix} m \times n & m \times m & m \times n & n \times n \end{matrix}$$

- When $B = Q^{-1}AP$ we call A and B are equivalent.
- Fact: Equivalent matrices have the same rank.
And the rank is the rank of the linear map the matrix encodes.

Corollary: If $T \in \mathcal{L}(V)$ and $\mathcal{E} = (e_1, \dots, e_n)$, $\mathcal{E}' = (e'_1, \dots, e'_n)$,
and $(e'_1, \dots, e'_n) = (e_1, \dots, e_n)P$

The matrix of T under $\mathcal{E}$ is A

" " of T under $\mathcal{E}'$ is B

Then $B = P^{-1}AP$

$$[T(v)]_{\mathcal{E}} = P[T(v)]_{\mathcal{E}'}$$

$$(e'_1, \dots, e'_n) = (e_1, \dots, e_n)P$$

↓ coord.

$$[v]_{\mathcal{E}} = P[v]_{\mathcal{E}'}$$

$$= PB[v]_{\mathcal{E}'} \quad \parallel$$

$$[T(v)]_{\mathcal{E}} = A[v]_{\mathcal{E}} = AP[v]_{\mathcal{E}'}$$

When $B = P^{-1}AP$ we say that A and B are similar. Note for $\mathcal{L}(V)$ the matrix is square: Only square matrices of the same size can be tested whether they are similar.

- $B = Q^{-1}AP$ (Equivalence) for $m \times n$ A & B

• $B = P^{-1}AP$ (Similarity) for $n \times n$ A & B .

► Two matrices are equivalent iff they have the same rank.

→ under some given basis.
?: For a matrix A (describing linear map $T: V \rightarrow W$)

Can we find alternative bases for V and W .

such that $B = Q^{-1}AP$ is as simple as possible?

Ans: Yes! In fact:

$$\exists P \& Q \text{ invertible: } Q^{-1}AP = \begin{matrix} r & n-r \\ \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \end{matrix}_{m \times n}$$

$$r = \text{Rank}(A) = \text{Rank}(T).$$

Rank Normal Form

?: For square matrices, A and B are similar
iff _____.

"Ans": There is no simple answer.

We will see a case where for certain matrices $P^{-1}AP = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$ (Diagonalization).

In general, the conditions are complicated and it depends on the field K .

Remark: If you are interested, search "Jordan Form"
"Frobenius Form" of similar matrices.

- Basis Extension Lemma.

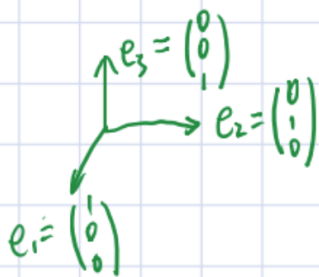
$(V, +, \cdot)$ vec. space. and $v_1, \dots, v_k$ Lin. indep.

Suppose $n = \dim V$ and $n > k$. ($n < \infty$)

Then we can find $v_{k+1}, v_{k+2}, \dots, v_n \in V$ Lin. indep.
and $v_1, \dots, v_k, v_{k+1}, \dots, v_n$ form a basis of V

Proof. is by an iterative argument.

Example: $\mathbb{R}^3$:


$$e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$
$$e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$
$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

- Invariant Subspace of a linear transformation

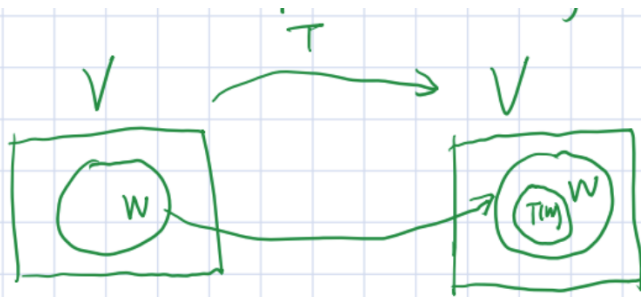
$T \in \mathcal{L}(V)$ ($T: V \rightarrow V$, linear)

A subspace $W \leq V$ is invariant if

$$T(W) = \{T(w) : w \in W\} \subseteq W.$$

$$V \xrightarrow{T} V$$

Clearly V is an invariant



Clearly V is an invariant subspace for any $T \in \mathcal{L}(V)$ and $\{0_V\}$ as well.

- Two important invariant subspaces of $T \in \mathcal{L}(V)$

- ARE SUBSPACES
- $\text{Im } T = \{T(v) : v \in V\}$ ← What's the output of T
 - $\text{ker } T = \{v \in V : T(v) = 0_V\}$ ← What's being lost under T

Claim: For any $T \in \mathcal{L}(V)$, $\text{Im } T$ and $\text{ker } T$ are invariant.

Proof: Take $w \in \text{Im } T \subseteq V$, apply T , $T(w) \in \text{Im } T$ by defn.

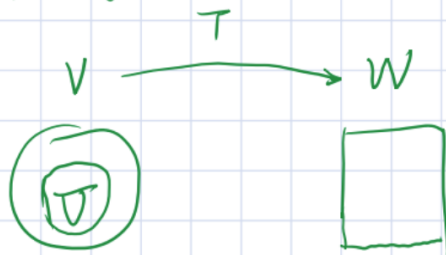
Take $u \in \text{ker } T$, then $T(u) = 0_V$.

Apply T : $T(T(u)) = T(0_V) = 0_V \Rightarrow T(u) \in \text{ker } T$.

$$T(0_V) = T(0_V + 0_V) = T(0_V) + T(0_V) \Rightarrow T(0_V) = 0_V$$

□

The above definition of $\text{Im } T$ and $\ker T$ goes also for linear maps $\mathcal{L}(V, W)$ but the concept of invariant subspaces only works for $\mathcal{L}(V)$.



" $T(U) \subseteq U$ " doesn't make sense since $T(U) \subseteq W \neq V$.

In this case, $\text{Im } T \subseteq W$ and $\ker T \subseteq V$.

- Rank, nullity and the rank-nullity Theorem.

- $\text{Rank}(T) := \dim \text{Im } T \leftarrow$ How many dimensions in W is the output of T ?

$\text{nullity}(T) := \dim \ker T \leftarrow$ How many dimensions in V is being mapped to 0_W by T ?

- Claim: (1) $T \in \mathcal{L}(V, W)$ is injective $\Leftrightarrow \ker T = \{0_V\}$

(2) $T \in \mathcal{L}(V, W)$ is $\overset{\text{one-to-one}}{\text{surjective}} \Leftrightarrow \overset{\text{onto}}{\text{Im } T} = W$

Proof: (2) is obvious.

(1) $\Rightarrow$: Suppose T injective and take $v \in \ker T$
 $T(v) = 0_W = T(0_V)$

By injectivity, $v = 0_V$

$\Leftarrow$: Suppose $\ker T = \{0_V\}$.

If $T(u) = T(v)$, by linearity:

$$T(u-v) = 0_W \Rightarrow u-v \in \ker T = \{0_V\}$$

This means $u = v$.

This proves T is injective.

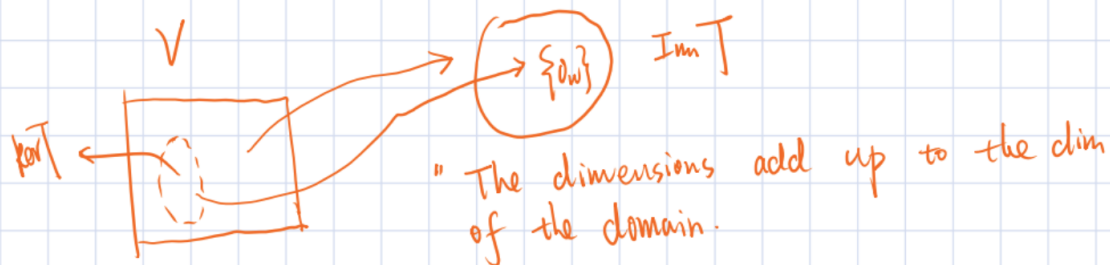
□

?: How are $\text{Rank}(T) := \dim \text{Im} T$ and $\text{nullity}(T) := \dim \ker T$ related to each other?

Ans: Rank-Nullity Theorem:

Let $T: V \rightarrow W$ Linear. ($\dim V < \infty$, $\dim W < \infty$)

Then $\dim V = \dim \text{Im} T + \dim \ker T$.



Equivalently: $\dim V = \text{Rank}(T) + \text{nullity}(T)$.

Proof: Let $u_1, \dots, u_k$ be a basis of $\ker T$ thus

$$\dim \ker T = \text{nullity}(T) = k.$$

$$u_1, \dots, u_k \in \ker T \subseteq V.$$

By Basis Extension we can find $v_1, \dots, v_r$ lin indep.

and $u_1, \dots, u_k, v_1, \dots, v_r$ is a basis for V .

$$\text{So } \dim V = k + \underline{r}.$$

We want to show that $T(v_1), \dots, T(v_r)$ forms a basis for $\text{Im} T$.

First: $T(v_1), \dots, T(v_r)$ span $\text{Im} T$.

Take any $w \in \text{Im} T$, by defn. $\exists v \in V: w = T(v)$.

Since $\underbrace{u_1, \dots, u_k}_{\in \ker T}, v_1, \dots, v_r$ is a basis for V

$$v = a_1 u_1 + \dots + a_k u_k + b_1 v_1 + \dots + b_r v_r \quad (*)$$

Apply T to both sides of $(*)$

$$w = T(v) = T(a_1 u_1 + \dots + a_k u_k + b_1 v_1 + \dots + b_r v_r)$$

$$= \underbrace{a_1 T(u_1) + \dots + a_k T(u_k)}_{0_W} + b_1 T(v_1) + \dots + b_r T(v_r)$$

$$= \sum_{j=1}^r b_j T(v_j)$$

Thus $\text{Im } T \subseteq \text{Span}(T(v_1), \dots, T(v_r))$

$\supseteq$: This is obvious since T is linear.

Therefore $\text{Im } T = \text{Span}(T(v_1), \dots, T(v_r))$.

Now. WTS. (= want to show) $T(v_1), \dots, T(v_r)$ lin. indep.

Suppose $c_1 T(v_1) + \dots + c_r T(v_r) = 0_W$

By linearity $0_W = T(c_1 v_1 + \dots + c_r v_r)$

This means $c_1 v_1 + \dots + c_r v_r \in \ker T$.

But recall that $u_1, \dots, u_k$ is a basis for $\ker T$

$$c_1 v_1 + \dots + c_r v_r = d_1 u_1 + \dots + d_k u_k.$$

$$\Rightarrow c_1 v_1 + \dots + c_r v_r - d_1 u_1 - \dots - d_k u_k = 0_V$$

Since $v_1, \dots, v_r, u_1, \dots, u_k$ is a basis $\checkmark \Rightarrow$ lin. indep.

$$\underbrace{c_1 = \dots = c_r}_{\triangle} = d_1 = \dots = d_k = 0_K$$

This shows that $T(v_1), \dots, T(v_r)$ is lin. indep.

Lin. indep + spans $\text{Im } T \Rightarrow T(v_1), \dots, T(v_r)$ is basis for $\text{Im } T$

$$\text{Rank}(T) = \dim \text{Im} T = \# \text{ basis elements} = r.$$

$$\Rightarrow \dim V = k + r = \dim \ker T + \dim \text{Im} T \\ = \text{nullity}(T) + \text{Rank}(T).$$

□

- Plan:
- ① Take basis for $\ker T$ in V
 - ② Extend it to a full basis of V
 - ③ Show that $\{T(\text{added basis elements})\}$ forms a basis of $\text{Im} T$
 - ④. Dimension counts follows.

Application:

Let $T \in \mathcal{L}(V, W)$ and $\dim V = \dim W$ ($\mathcal{L}(V)$ is a special case)

Then TFAE (the following are equivalent)

(1) $\ker T = \{0_V\}$

(2) T injective.

(3) T surjective

(4) T bijective.

→ If $\ker T = \{0_V\}$, then by R-N:

$$\dim V = \dim \text{Im} T + \underbrace{\dim \ker T}_0 = \dim \text{Im} T$$

||

$\dim W$. This means $\text{Im} T = W$

because $\text{Im} T \leq W$

Then T is surjective

Surjective $\Rightarrow \dim W = \dim \text{Im} T$ plug into R-N.

$$\Rightarrow \dim \ker T = 0 \Rightarrow \ker T = \{0_V\}$$

= Rank of a matrix:

• Take $A \in K^{m \times n}$, then A naturally gives a linear map

$$T_A : K^n \longrightarrow K^m$$

$$x \longmapsto Ax$$

• What is Ax ?

$$Ax = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + \dots + a_{1n}x_n \\ a_{21}x_1 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n \end{pmatrix}$$
$$= x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \dots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}$$

$$\text{Then } \text{Im } T_A = \{ Ax : x \in K^n \}$$

$$= \left\{ x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + \dots + x_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} : x_1, \dots, x_n \in K \right\}$$

$$= \text{Span} \left(\begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix}, \dots, \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} \right)$$

$$:= \text{Col}(A)$$

Small note: $\dim \text{Col}(A)$ is technically speaking the column rank of A .

There is an analogous concept: row rank

$\dim \text{Row}(A)$ which is the dimension of the Span of the rows of A

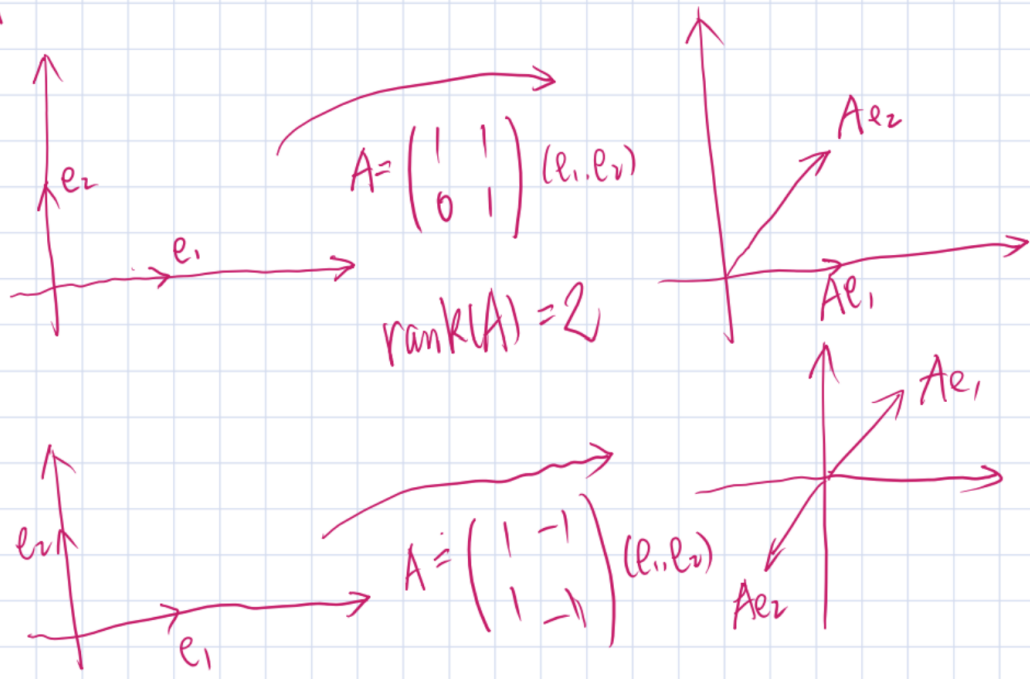
K_n

$$\dim \text{Col}(A) = \dim \text{Row}(A)$$

Take: $\mathbb{R}^{2 \times 3} : \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 4 \end{pmatrix}$ for example.

- Rank tells us how many lin indep. rows / columns there are in a matrix. Viewed as a linear map: how many indep. directions are being output.

Example: $\mathbb{L}(\mathbb{R}^2)$



• Sum and Direct Sum of Subspaces.

Take $V_1, \dots, V_K : V_i \subseteq V$ the sum:

$$V_1 + V_2 + \dots + V_K = \bigoplus_{i=1}^K V_i = \{v_1 + v_2 + \dots + v_K : v_i \in V_i, \forall i=1, \dots, K\}$$

If $v \in \bigoplus_{i=1}^K V_i$, then $v = v_1 + v_2 + \dots + v_K$ for some $v_i \in V_i, \forall i$.

If this representation is unique:

If $v \in \bigoplus_{i=1}^K V_i$ then $\exists! v_1, \dots, v_K : v = v_1 + v_2 + \dots + v_K$

Then we call this sum a direct sum.

Denote it as $V_1 \oplus V_2 \oplus \dots \oplus V_K = \bigoplus_{i=1}^K V_i$.

Example: $\mathbb{R}^3 = \text{Span}\left\{ \begin{matrix} e_1 \\ e_2 \\ e_3 \end{matrix} \right\}$

For any $v \in \mathbb{R}^3 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 e_1 + x_2 e_2 + x_3 e_3$

$$\mathbb{R}^3 = \text{Span}\{e_1\} \oplus \text{Span}\{e_2\} \oplus \text{Span}\{e_3\}.$$



But $\mathbb{R}^3 = \text{Span}\{e_1, e_2\} + \text{Span}\{e_2, e_3\}$
(Check this).

Claim: This sum is not direct.

$$\begin{aligned} \text{Let's take } e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} &= \frac{1}{2} \left[\underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}_{\text{Span}\{e_1, e_2\}} + \underbrace{\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}}_{\text{Span}\{e_2, e_3\}} \right] + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \frac{1}{2} \left[\underbrace{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}}_{\text{Span}\{e_1, e_2\}} + \underbrace{\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}}_{\text{Span}\{e_2, e_3\}} \right] \end{aligned}$$

So the rep. of e_2 is NOT UNIQUE.

So the sum is not direct by def.

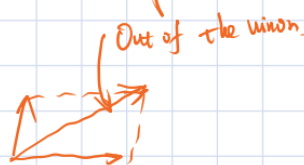
- Why sum?

Sum of vector spaces forms a vec. space.

(Check!)

Union of vec. spaces is usually NOT a vec. space.

Counterexample in $\mathbb{R}^2$: $\text{Span}\{e_1\} \cup \text{Span}\{e_2\}$.



NOT A Vec. Space!

- Determinants.

! : Determinants only exists for square matrices.

What is the determinant?

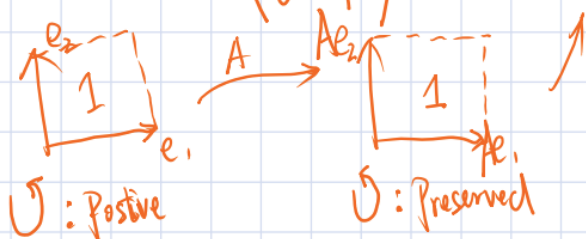
Take $T \in \mathcal{L}(\mathbb{R}^n)$.

$\det(T)$ = signal volume $T([0,1]^n)$.

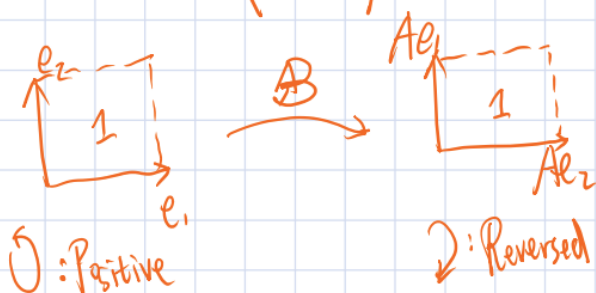
We've seen det. for 2×2 matrices.

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

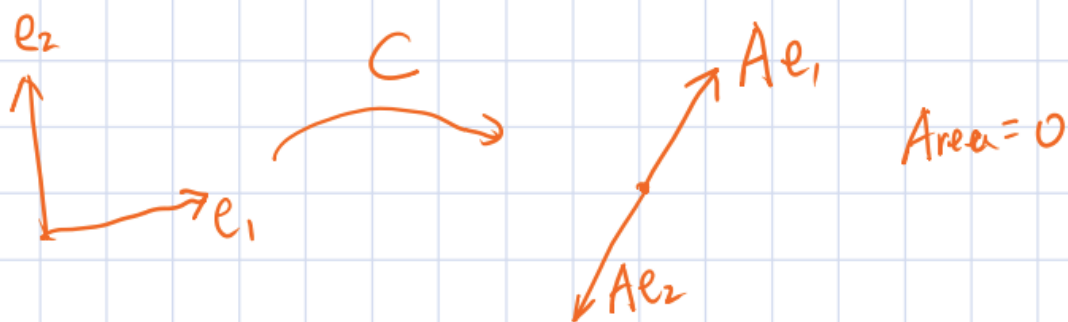
Consider: $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\det(A) = 1$



$B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\det(A) = -1$



$$C = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}, \quad \det(C) = 0$$



Note that C has lin. dep. columns.

$$1 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- What's generally true is that if A $n \times n$.
 $\text{rank}(A) = n \Leftrightarrow \det(A) \neq 0$
 $\Leftrightarrow A$ invertible.

Take e_i as the standard basis vector

$\rightarrow Ae_i = i^{\text{th}}$ column of A (Verify)

$$A(e_1, \dots, e_n) = A I_n = A$$

If curious / unclear. Watch 3blue1brown on linear algebra.

Today : - Rank , Nullity , R-N Thm.

- Change of basis for $T \in \mathcal{L}(V, W)$.

- Equivalence & Similarity

- Sum and direct sum

- Determinants

Coming Next : Diagonalization & Inner product Spaces.